

INT335: DESIGN THINKING

180 Concept-Building MCQs & Exam Practice Bank

Target Curriculum:

B.Tech Computer Science & Engineering | Semester 3

Lovely Professional University

Question Bank Features:

- **Complete 6-Unit Coverage:** 30 easy, verified MCQs per Unit (180 Total).
- **Comprehensive Solutions:** Every question includes correct answer key and pedagogical rationale.
- **Aligned with Syllabus:** Covers cognitive models, user empathy, SCAMPER, wireframes, testing heuristics, and NABC pitching.

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CHAPTER 1

Unit 1: Foundations of Learning, Creativity and Design Thinking

Practice Question Set (30 MCQs)

Q.1 What is the fundamental starting point of Design Thinking?

- (A) Cutting-edge technology
- (B) Market profit maximization
- (C) Human needs and user empathy
- (D) Government regulations

Q.2 Who coined the term "Wicked Problems" in 1973 to describe complex, ill-defined social challenges?

- (A) Herbert Simon and Peter Rowe
- (B) Horst Rittel and Melvin Webber
- (C) David Kelley and Tim Brown
- (D) Karl Duncker and David Kolb

Q.3 Which of the following is NOT one of IDEO's three overlapping lenses of innovation?

- (A) Desirability
- (B) Feasibility
- (C) Viability
- (D) Predictability

Q.4 What type of thinking focuses on generating a broad variety of new and novel ideas?

- (A) Convergent thinking
- (B) Divergent thinking
- (C) Critical thinking
- (D) Deductive reasoning

Q.5 In cognitive learning, why is active retrieval superior to passive re-reading?

- (A) It takes less time
- (B) It strengthens neural memory traces and recall pathways
- (C) It avoids critical analysis
- (D) It relies exclusively on short-term memory

- Q.6** Which brain mode is characterized by relaxed attention and subconscious incubation of ideas?
- (A) Focused mode
 - (B) Diffuse mode
 - (C) Analytical mode
 - (D) Reflexive mode
- Q.7** What is "Functional Fixedness"?
- (A) Inability to change hardware components
 - (B) A cognitive bias where an object is seen only in terms of its traditional use
 - (C) The capacity to repair broken tools
 - (D) A systematic software testing methodology
- Q.8** Karl Duncker's famous 1945 experiment demonstrating functional fixedness is known as the:
- (A) Box and String Test
 - (B) Candle Problem
 - (C) Nine Dots Puzzle
 - (D) Tower of Hanoi
- Q.9** Which sequence accurately represents the four stages of Kolb's Experiential Learning Cycle?
- (A) Remember → Understand → Apply → Analyze
 - (B) Concrete Experience → Reflective Observation → Abstract Conceptualization → Active Experimentation
 - (C) Ideate → Define → Prototype → Test
 - (D) Plan → Do → Check → Act
- Q.10** What does the acronym VARK stand for in learning styles?
- (A) Verbal, Analytical, Reasoning, Knowledge
 - (B) Visual, Aural, Read/Write, Kinesthetic
 - (C) Vision, Auditory, Reality, Kinetics
 - (D) Variable, Action, Reflex, Kinetic
- Q.11** What is the highest cognitive level in Bloom's Revised Taxonomy?
- (A) Analyze
 - (B) Evaluate
 - (C) Create
 - (D) Apply

Q.12 What distinguishes an "Innovation" from an "Invention"?

- (A) Innovation does not require technology
- (B) Innovation involves successful implementation and real-world adoption of value
- (C) Inventions are always cheaper than innovations
- (D) Innovation is always patented

Q.13 In which book did Herbert Simon state that design is "transforming existing conditions into preferred ones"?

- (A) The Sciences of the Artificial (1969)
- (B) Design Thinking (1987)
- (C) The Design of Everyday Things (1988)
- (D) Change by Design (2009)

Q.14 The German design school established in 1919 that united art, craft, and functional design was:

- (A) Bauhaus
- (B) Stanford d.school
- (C) Ulm School
- (D) Royal College of Art

Q.15 Who authored the 1987 book titled *Design Thinking*, focusing on architectural reasoning?

- (A) Tim Brown
- (B) Peter Rowe
- (C) Don Norman
- (D) Alex Osborn

Q.16 In which year was the Stanford d.school (Hasso Plattner Institute of Design) founded?

- (A) 1991
- (B) 1999
- (C) 2005
- (D) 2015

Q.17 What are the 5 iterative modes of the Stanford d.school Design Thinking process?

- (A) Listen, Think, Build, Deploy, Maintain
- (B) Empathize, Define, Ideate, Prototype, Test
- (C) Discover, Design, Develop, Deliver, Debrief
- (D) Plan, Draft, Review, Publish, Archive

Q.18 What are the three phases of the IDEO Human-Centred Design framework?

- (A) Input, Process, Output
- (B) Inspiration, Ideation, Implementation
- (C) Investigation, Iteration, Integration
- (D) Initiation, Incubation, Impact

Q.19 Which product design success featured thick, comfortable handles originally developed for arthritis sufferers?

- (A) Juicero
- (B) OXO Good Grips
- (C) Apple iPhone
- (D) Ford Edsel

Q.20 What made Kenya's M-Pesa a global design thinking success in 2007?

- (A) High-end 4G smartphone apps
- (B) Simple SMS-based money transfers supported by local agent networks
- (C) Zero-fee luxury banking lounges
- (D) Crypto-currency smart contracts

Q.21 Why is the Jaipur Foot considered a human-centred design triumph in India?

- (A) It uses carbon fiber imported from Germany
- (B) It is customized for local customs like barefoot walking, farming, and squatting at low cost
- (C) It includes built-in electronic tracking
- (D) It was designed without consulting disabled users

Q.22 Why did the Juicero press (2017) fail commercially?

- (A) It was prone to electrical fires
- (B) Users discovered the expensive proprietary juice bags could easily be squeezed by hand
- (C) It lacked Wi-Fi connectivity
- (D) It was banned by FDA regulations

Q.23 Why did the consumer edition of Google Glass face severe market backlash?

- (A) Poor battery life only
- (B) Privacy concerns regarding hidden recording, social awkwardness, and unclear everyday value
- (C) High weight causing physical injury
- (D) Inability to connect to the internet

Q.24 Deliberate practice is characterized by which of the following?

- (A) Mindless repetition without feedback
- (B) Focused effort on weaknesses combined with immediate corrective feedback
- (C) Multitasking during learning
- (D) Memorizing answers without understanding

Q.25 Metacognition is best defined as:

- (A) The speed of brain processing
- (B) Thinking about and regulating one's own learning and thinking processes
- (C) Emotional intelligence in social groups
- (D) Telepathic connection between team members

Q.26 What is the capacity of the brain to reorganize itself by forming new neural connections throughout life?

- (A) Neurogenesis
- (B) Neuroplasticity
- (C) Functional fixedness
- (D) Synaptic pruning

Q.27 In creative problem solving, what role does a failed prototype play?

- (A) Total loss of project budget
- (B) Cheap and rapid discovery of flawed assumptions
- (C) Proof that the design team lacks talent
- (D) Reason to permanently abandon the project

Q.28 Which thinking mode applies logic, filters criteria, and evaluates constraints to choose the best solution?

- (A) Divergent thinking
- (B) Convergent thinking
- (C) Lateral thinking
- (D) Associative thinking

Q.29 What is a "Mental Set"?

- (A) A positive mindset before starting work
- (B) The tendency to rely rigidly on previously successful methods even when simpler approaches exist
- (C) A collection of brainstorming tools
- (D) Psychological readiness to accept feedback

Q.30 Which of the following is a primary countermeasure to Functional Fixedness?

- (A) Sticking to the user manual
- (B) Reframing the problem and deconstructing objects into fundamental properties
- (C) Increasing project deadlines
- (D) Asking for managerial approval

Answer Key & Step-by-Step Explanations

Unit 1: Quick Answer Key

Q1 C	Q2 B	Q3 D	Q4 B	Q5 B	Q6 B	Q7 B	Q8 B	Q9 B	Q10 B
Q11 C	Q12 B	Q13 A	Q14 A	Q15 B	Q16 C	Q17 B	Q18 B	Q19 B	Q20 B
Q21 B	Q22 B	Q23 B	Q24 B	Q25 B	Q26 B	Q27 B	Q28 B	Q29 B	Q30 B

Detailed Pedagogical Explanations

1. **[Q1: Option C]** Design Thinking always begins with user empathy and understanding human needs.
2. **[Q2: Option B]** Horst Rittel and Melvin Webber coined "Wicked Problems" in 1973.
3. **[Q3: Option D]** The three lenses are Desirability (people), Feasibility (technology), and Viability (business).
4. **[Q4: Option B]** Divergent thinking expands the solution space by generating numerous diverse ideas.
5. **[Q5: Option B]** Active recall actively engages neural memory networks, making retrieval durable.
6. **[Q6: Option B]** Diffuse mode allows unconscious incubation and wide-ranging connections across the brain.
7. **[Q7: Option B]** Functional fixedness limits a person to seeing an object only in its traditional role.
8. **[Q8: Option B]** Duncker's candle problem tested using a matchbox as a shelf rather than just a container.
9. **[Q9: Option B]** Kolb's cycle moves: Concrete Experience → Reflective Observation → Abstract Conceptualization → Active Experimentation.
10. **[Q10: Option B]** VARK represents Visual, Aural, Read/Write, and Kinesthetic learning modalities.
11. **[Q11: Option C]** In Bloom's revised taxonomy, 'Create' is at the very apex of cognitive skills.
12. **[Q12: Option B]** An invention is a novel device/process; an innovation is successfully implemented to deliver value.
13. **[Q13: Option A]** Herbert Simon wrote *The Sciences of the Artificial* in 1969.
14. **[Q14: Option A]** Bauhaus was founded in 1919 in Germany, integrating craft, art, and industrial design.

15. [Q15: Option B] Peter Rowe published *Design Thinking* in 1987.
16. [Q16: Option C] The Stanford d.school was founded in 2005 by David Kelley.
17. [Q17: Option B] Stanford d.school's 5 stages: Empathize, Define, Ideate, Prototype, and Test.
18. [Q18: Option B] IDEO's 3I framework consists of Inspiration, Ideation, and Implementation.
19. [Q19: Option B] OXO Good Grips created ergonomic kitchen tools that became a classic universal design hit.
20. [Q20: Option B] M-Pesa utilized basic SMS technology and local agents to serve unbanked populations.
21. [Q21: Option B] Jaipur Foot adapted prosthetics to Indian cultural practices (barefoot walking, squatting).
22. [Q22: Option B] Juicero was over-engineered and solved a non-problem since juice packs were hand-squeezeable.
23. [Q23: Option B] Google Glass faced severe social resistance due to perceived privacy violations and lack of clear consumer utility.
24. [Q24: Option B] Deliberate practice requires focused concentration on errors with immediate feedback.
25. [Q25: Option B] Metacognition is conscious awareness and control over one's own thinking.
26. [Q26: Option B] Neuroplasticity is the brain's ability to rewire its neural pathways through experience and learning.
27. [Q27: Option B] In design thinking, early prototype failure is an inexpensive source of validated learning.
28. [Q28: Option B] Convergent thinking filters, evaluates, and selects from generated possibilities.
29. [Q29: Option B] A mental set is a cognitive habit of using familiar problem-solving methods rigidly.
30. [Q30: Option B] Reframing and breaking objects down into raw properties breaks functional fixedness.

CHAPTER 2

Unit 2: Empathy, Observation and Problem Identification

Practice Question Set (30 MCQs)

Q.1 What is empathy in the context of Design Thinking?

- (A) Feeling pity or sympathy for users
- (B) The deep understanding of user feelings, motivations, behaviors, and unmet needs
- (C) Selling products using emotional advertisements
- (D) Guessing what users want based on designer intuition

Q.2 Which statement correctly differentiates a user "need" from a "solution"?

- (A) "Users need an iPhone app" is a need; "Faster communication" is a solution
- (B) "Users need timely bus arrival info" is a need; "A mobile GPS tracker" is a solution
- (C) Needs and solutions are identical terms in design thinking
- (D) A solution is always identified before the user need

Q.3 What is "Computational Empathy" in software engineering?

- (A) Giving computers feelings and consciousness
- (B) Designing systems that adapt gracefully to user context, errors, and emotional states
- (C) Calculating mathematical probabilities of human emotions
- (D) Replacing human customer support agents with chatbots

Q.4 Qualitative user research primarily aims to answer which question?

- (A) "How many users clicked the button?"
- (B) "What is the average transaction value?"
- (C) "Why and how do users experience this problem?"
- (D) "What percentage of users are on iOS?"

Q.5 What is "Data Triangulation" in user research?

- (A) Surveying exactly three users
- (B) Cross-verifying findings using multiple independent research methods and data sources
- (C) Arranging interview transcripts in triangular shapes
- (D) Using three different chart types in a presentation

Q.6 What does the "E" stand for in the AEIOU observation framework?

- (A) Emotion
- (B) Environment
- (C) Evaluation
- (D) Efficiency

Q.7 In the AEIOU framework, "Interactions" refers to:

- (A) Building relationships between project managers
- (B) Exchanges between person-to-person, person-to-device, or person-to-object
- (C) Social media marketing campaigns
- (D) Network server communications

Q.8 In user interviews, why are open-ended questions preferred over closed-ended questions?

- (A) They elicit rich stories, past behaviors, and detailed context rather than simple "Yes/No"
- (B) They take much less time to administer
- (C) They are easier to grade mathematically
- (D) They prevent participants from talking too much

Q.9 Which of the following is an example of a LEADING question that should be AVOIDED?

- (A) "Tell me about the last time you booked a doctor's appointment."
- (B) "What challenges did you face during your recent online order?"
- (C) "Don't you agree that our new checkout screen is very fast and helpful?"
- (D) "How did you feel when the payment failed?"

Q.10 The "Laddering" or "5 Whys" technique is primarily used to:

- (A) Count the number of steps in a software flow
- (B) Uncover deep underlying root causes and emotional drivers behind a surface behavior
- (C) Grade user intelligence
- (D) Prolong the duration of an interview session

Q.11 What is an "Explicit Need"?

- (A) An unspoken subconscious desire
- (B) A directly verbalized and clearly stated user requirement
- (C) A requirement that violates government regulations
- (D) An engineering specification hidden in code

Q.12 What is an "Implicit (Latent) Need"?

- (A) An unstated expectation inferred through observing user workarounds, body language, and hesitation
- (B) A feature written in the user manual
- (C) A bug reported on an official feedback form
- (D) A need that is impossible to satisfy

Q.13 When a user has to enter the exact same shipping address across 4 different checkout screens, what type of pain point is this?

- (A) Financial pain point
- (B) Time / Process pain point
- (C) Aesthetic pain point
- (D) Environmental pain point

Q.14 Hidden delivery charges revealed only on the final payment page create which type of user pain point?

- (A) Process pain
- (B) Financial pain
- (C) Technical pain
- (D) Physical pain

Q.15 A User Persona in Design Thinking should be created based on:

- (A) A designer's personal assumptions and stereotypes
- (B) Empirical data, user research patterns, behaviors, and motivations
- (C) Purely demographic averages (age, gender, income) without behavioral context
- (D) Fictional characters from popular movies

Q.16 What is the primary difference between a Primary Persona and a Secondary Persona?

- (A) Primary personas are real people; secondary personas are robots
- (B) Primary persona is the core target user whose needs dictate system design; secondary needs must not compromise primary goals
- (C) Secondary personas pay for the software; primary personas only use it
- (D) Primary personas are older adults; secondary personas are youth

Q.17 Which of the following is NOT one of the four main quadrants of a standard Empathy Map?

- (A) Says
- (B) Thinks
- (C) Codes
- (D) Feels

Q.18 In an Empathy Map, where do observable actions, habits, and physical workarounds belong?

- (A) Thinks
- (B) Feels
- (C) Does
- (D) Says

Q.19 Why is it important to distinguish between "Says" and "Does" in an Empathy Map?

- (A) Users often state what is socially desirable but behave differently in reality
- (B) "Says" is always accurate while "Does" is misleading
- (C) Designers only care about what users say
- (D) "Does" only applies to hardware devices

Q.20 What is a "Point-of-View" (POV) statement in the Define stage?

- (A) A legal copyright statement
- (B) A meaningful problem statement combining User, Need, and Insight
- (C) The marketing slogan of the company
- (D) A technical bug report

Q.21 How do design teams convert a POV problem statement into a launchpad for creative brainstorming?

- (A) By writing software requirement specifications (SRS)
- (B) By creating "How Might We" (HMW) questions
- (C) By immediately purchasing off-the-shelf software
- (D) By calculating revenue forecasts

Q.22 What makes a "How Might We" (HMW) question effective?

- (A) It is so narrow that it includes the exact solution
- (B) It is broad enough to allow creative exploration, yet narrow enough to provide clear focus
- (C) It can be answered with a simple "Yes" or "No"
- (D) It focuses exclusively on technology costs

Q.23 In the AEIOU framework, a handwritten cheat-sheet taped to a computer monitor is categorized as an:

- (A) Activity
- (B) Environment
- (C) Object
- (D) User

Q.24 During observational field research, what is the best practice for researchers?

- (A) Immediately intervene and correct the user when they make an error
- (B) Observe silently, timestamp events, and record actual behavior without judgment
- (C) Pitch new product ideas to the user while they work
- (D) Ignore the environment and only look at the computer screen

Q.25 What is the primary ethical responsibility of a design researcher during interviews?

- (A) Guaranteeing that participants will receive free products
- (B) Obtaining informed consent, ensuring data privacy, and allowing withdrawal at any time
- (C) Recording audio without informing the user to ensure natural responses
- (D) Convincing the participant that the current product is perfect

Q.26 Which of the following is a classic indicator of an unmet user need during observation?

- (A) The user completes the task effortlessly in 5 seconds
- (B) The user creates an improvised workaround (e.g., using personal spreadsheets or paper notes)
- (C) The user follows the official documentation without hesitation
- (D) The user smiles and praises the software

Q.27 How does quantitative research support qualitative research in the empathy phase?

- (A) It replaces qualitative research completely
- (B) It validates how widespread a discovered qualitative pain point is across a broader population
- (C) It generates emotional empathy maps
- (D) It eliminates the need to interview real users

Q.28 An actionable design insight is best described as:

- (A) A raw user quote copied directly into a report
- (B) A statistical graph showing website traffic
- (C) A deep, non-obvious realization that connects user behavior to underlying human motivations
- (D) A list of software features requested by management

Q.29 What type of pain point occurs when a user feels anxious and embarrassed because an ATM display exposes their account balance to everyone in line?

- (A) Physical pain
- (B) Emotional / Privacy pain
- (C) Financial pain
- (D) Operational pain

Q.30 Which of the following defines a Target Archetype?

- (A) A strict mathematical demographic model
- (B) A generalized behavioral profile capturing shared goals, mental models, and usage contexts
- (C) A 3D CAD model of an average human body
- (D) The software architecture blueprint

Answer Key & Step-by-Step Explanations

Unit 2: Quick Answer Key

Q1 B	Q2 B	Q3 B	Q4 C	Q5 B	Q6 B	Q7 B	Q8 A	Q9 C	Q10 B
Q11 B	Q12 A	Q13 B	Q14 B	Q15 B	Q16 B	Q17 C	Q18 C	Q19 A	Q20 B
Q21 B	Q22 B	Q23 C	Q24 B	Q25 B	Q26 B	Q27 B	Q28 C	Q29 B	Q30 B

Detailed Pedagogical Explanations

1. **[Q1: Option B]** Empathy is understanding the user's emotions, experiences, and implicit needs from their perspective.
2. **[Q2: Option B]** Needs are underlying goals/outcomes; solutions are specific implementations.
3. **[Q3: Option B]** Computational empathy designs software that adapts to human context, cognitive load, and emotions.
4. **[Q4: Option C]** Qualitative research explores the rich "Why" and "How" behind human behavior.
5. **[Q5: Option B]** Triangulation compares multiple distinct research methods to validate findings.
6. **[Q6: Option B]** In AEIOU: Activities, Environments, Interactions, Objects, Users.
7. **[Q7: Option B]** Interactions cover exchanges between humans, devices, and physical objects.
8. **[Q8: Option A]** Open-ended questions allow participants to share realistic stories and unprompted context.
9. **[Q9: Option C]** Leading questions nudge the participant toward a desired positive answer and introduce heavy bias.
10. **[Q10: Option B]** Laddering (5 Whys) drills down from superficial actions to root motivations.
11. **[Q11: Option B]** Explicit needs are directly stated and clearly communicated by users.
12. **[Q12: Option A]** Implicit needs are unstated desires uncovered through observations and workarounds.
13. **[Q13: Option B]** Re-entering identical information is a classic time and workflow process friction.
14. **[Q14: Option B]** Unexpected costs and hidden pricing friction represent financial pain points.

15. **[Q15: Option B]** Personas must be grounded in real behavioral research data, not designer stereotypes.
16. **[Q16: Option B]** Primary personas represent the core user whose needs must be satisfied first without compromise.
17. **[Q17: Option C]** Empathy map quadrants are Says, Thinks, Does, and Feels (plus Pains & Gains).
18. **[Q18: Option C]** "Does" captures observable physical actions, workflows, and behavioral adaptations.
19. **[Q19: Option A]** People often say what sounds good or polite, but their actual behavior reveals true needs.
20. **[Q20: Option B]** A POV statement synthesizes User + Need + Surprising Insight.
21. **[Q21: Option B]** "How Might We" questions reframe problem insights into constructive ideation prompts.
22. **[Q22: Option B]** Good HMWs strike a balance: broad enough for variety, narrow enough for actionable focus.
23. **[Q23: Option C]** Physical artefacts, cheat-sheets, and improvised tools fall under 'Objects'.
24. **[Q24: Option B]** Neutral observation without interrupting or correcting yields authentic behavioral data.
25. **[Q25: Option B]** Respecting privacy, gaining informed consent, and protecting participants are ethical imperatives.
26. **[Q26: Option B]** Workarounds signal that existing official tools fail to satisfy a crucial user need.
27. **[Q27: Option B]** Quantitative data tests the scale and frequency of qualitative insights.
28. **[Q28: Option C]** Insights go beyond surface facts to explain the underlying "Why".
29. **[Q29: Option B]** Anxiety, vulnerability, and embarrassment are emotional/psychological pain points.
30. **[Q30: Option B]** Archetypes represent behavioral patterns and shared mental models across user groups.

CHAPTER 3

Unit 3: Ideation and Creative Problem Solving

Practice Question Set (30 MCQs)

Q.1 What is the primary objective of the Ideation stage in Design Thinking?

- (A) Writing production code immediately
- (B) Transitioning from problem definition to generating a broad range of potential solutions
- (C) Finalizing the product marketing budget
- (D) Performing financial audits

Q.2 Which golden rule must be maintained during the initial phase of idea generation?

- (A) Evaluate the financial cost of every idea immediately
- (B) Defer judgment and separate idea generation from idea evaluation
- (C) Allow only senior managers to speak
- (D) Reject any idea that cannot be built within one week

Q.3 In creativity assessment, what does "Fluency" measure?

- (A) How beautifully an idea is drawn
- (B) The total number of ideas generated within a given timeframe
- (C) The grammar and vocabulary used in the pitch
- (D) The technical complexity of the code

Q.4 In creativity assessment, what does "Flexibility" measure?

- (A) The physical flexibility of prototype materials
- (B) The number of different categories or domains represented among the ideas
- (C) The willingness of a team to work overtime
- (D) The adaptability of software servers

Q.5 Who established the foundational rules of modern Brainstorming in 1953?

- (A) Tim Brown
- (B) Alex Osborn
- (C) Karl Duncker
- (D) Steve Jobs

Q.6 Which improv phrase is widely used during brainstorming to build constructively on colleagues' ideas?

- (A) "No, but..."
- (B) "Yes, and..."
- (C) "Why would that work?"
- (D) "That is too expensive"

Q.7 What is "Brainwriting" (such as the 6-3-5 method)?

- (A) Writing software code directly in a notebook
- (B) A silent idea-generation technique where participants write ideas on paper and pass them around to build upon
- (C) Taking notes during a university lecture
- (D) Drawing mind maps individually without sharing

Q.8 In the SCAMPER framework, what does the letter 'S' represent?

- (A) Simplify
- (B) Substitute
- (C) Systematize
- (D) Scale

Q.9 In the SCAMPER framework, replacing paper bus tickets with a mobile QR code scan is an example of:

- (A) Put to another use
- (B) Substitute
- (C) Reverse
- (D) Magnify

Q.10 In SCAMPER, combining a luggage trolley with an electric scooter mechanism is an example of:

- (A) Combine
- (B) Eliminate
- (C) Reverse
- (D) Substitute

Q.11 In SCAMPER, what does 'E' stand for?

- (A) Expand
- (B) Evaluate
- (C) Eliminate

(D) Energize

Q.12 In SCAMPER, taking a self-checkout system where customers scan goods themselves instead of cashiers scanning them represents:

(A) Adapt

(B) Reverse / Rearrange

(C) Minify

(D) Substitute

Q.13 How does a Mind Map structure ideas?

(A) In a linear alphabetical list

(B) Radially outward from a central problem keyword into branching associative themes

(C) In a chronological timeline from left to right

(D) In a 2x2 mathematical matrix

Q.14 What is "Reverse Thinking" (Inversion / Worst Possible Idea)?

(A) Reading a design brief backwards

(B) Intentionally brainstorming ways to cause or worsen the problem, then inverting those ideas into solutions

(C) Copying a competitor's product in reverse

(D) Writing text from right to left

Q.15 What is the main purpose of "Affinity Diagramming"?

(A) Grouping large volumes of scattered ideas into natural thematic clusters based on relationships

(B) Calculating weighted mathematical scores

(C) Selecting one single winning idea randomly

(D) Testing prototype durability

Q.16 What is "Concept Bundling"?

(A) Selling multiple unsold products at a discount

(B) Packaging several complementary features into a coherent, high-value solution concept

(C) Archiving rejected ideas into zip files

(D) Merging two competing design teams

Q.17 Which of the following is a classic two-stage idea selection process?

(A) Blind voting → Random selection

(B) Initial coarse filter (feasibility/constraints) → Deep comparative evaluation matrix

(C) Manager selection → Customer survey

(D) Lowest cost → Immediate launch

Q.18 In a Weighted Screening Matrix, if Criterion A has a weight of 0.6 and Criterion B has a weight of 0.4, an idea scoring 5 on A and 2 on B gets a total score of:

(A) 7.0

(B) 3.8

(C) 4.2

(D) 2.9

Q.19 In an Impact-Effort Matrix, ideas that offer HIGH IMPACT with LOW EFFORT are classified as:

(A) Major Projects

(B) Quick Wins

(C) Thankless Tasks

(D) Fill-ins

Q.20 In an Impact-Effort Matrix, ideas that require HIGH EFFORT but yield LOW IMPACT should generally be:

(A) Implemented immediately

(B) Given the highest budget

(C) Dropped, avoided, or deprioritized

(D) Assigned to junior designers

Q.21 What is "Dot Voting" in an ideation workshop?

(A) A method where participants place colored adhesive dots next to their favorite ideas to visually map group consensus

(B) Drawing dotted lines across wireframes

(C) Writing punctuation marks on sticky notes

(D) Voting by sending text messages

Q.22 In the Double Diamond design model, when does ideation take place?

(A) The first diamond (Discover)

(B) The first diamond (Define)

(C) The opening of the second diamond (Develop / Diverge on solutions)

(D) After product launch

Q.23 Why is it dangerous to converge on the very first idea that comes to mind during brainstorming?

(A) The first idea is always illegal

- (B) First ideas are typically obvious, conventional, and lack deeper exploratory value
- (C) First ideas cannot be prototyped
- (D) Brainstorming rules require generating at least 500 ideas

Q.24 What is the role of a facilitator during a team brainstorming session?

- (A) Dictate the best solutions to the team
- (B) Enforce time limits, maintain energy, encourage quiet members, and uphold brainstorming rules
- (C) Criticize unfeasible suggestions immediately
- (D) Write all ideas without allowing team members to participate

Q.25 Which SCAMPER prompt is being used when a smartphone manufacturer removes the physical 3.5mm headphone jack to create more internal battery space?

- (A) Modify
- (B) Eliminate
- (C) Adapt
- (D) Put to another use

Q.26 What does "Originality" represent in idea generation?

- (A) How quickly an idea was conceived
- (B) The statistical rarity and unconventional nature of an idea compared to common standards
- (C) The patent registration number
- (D) The cost required to manufacture the idea

Q.27 When evaluating ideas, why is "Desirability" assessed before "Feasibility"?

- (A) Engineers do not work until marketing finishes
- (B) There is no point engineering a technically feasible system that nobody actually wants or needs
- (C) Feasibility is completely irrelevant in modern design
- (D) Desirability is always cheaper than feasibility

Q.28 What should a design team do with interesting ideas that are not selected for the current MVP?

- (A) Delete them permanently from all records
- (B) Archive them in an "Idea Parking Lot" / backlog for future iterations or reference
- (C) Penalize the team members who created them
- (D) Sell them to competitors

Q.29 What is "Production Blocking" in traditional verbal group brainstorming?

- (A) Running out of sticky notes and markers

- (B) A phenomenon where participants forget or suppress their ideas while waiting for others to stop speaking
- (C) Factory manufacturing machine breakdown
- (D) Managers canceling a brainstorming workshop

Q.30 Which ideation technique is specifically designed to eliminate Production Blocking and evaluation apprehension?

- (A) Open verbal round-robin
- (B) Silent Brainwriting (6-3-5)
- (C) Public debate
- (D) Impromptu pitching

Answer Key & Step-by-Step Explanations

Unit 3: Quick Answer Key

Q1 B	Q2 B	Q3 B	Q4 B	Q5 B	Q6 B	Q7 B	Q8 B	Q9 B	Q10 A
Q11 C	Q12 B	Q13 B	Q14 B	Q15 A	Q16 B	Q17 B	Q18 B	Q19 B	Q20 C
Q21 A	Q22 C	Q23 B	Q24 B	Q25 B	Q26 B	Q27 B	Q28 B	Q29 B	Q30 B

Detailed Pedagogical Explanations

1. **[Q1: Option B]** Ideation transitions from problem framing to creating diverse candidate solutions.
2. **[Q2: Option B]** Deferring judgment ensures psychological safety and wide exploratory breadth.
3. **[Q3: Option B]** Fluency is the sheer numerical volume of ideas produced.
4. **[Q4: Option B]** Flexibility measures the diversity of categories and perspectives represented.
5. **[Q5: Option B]** Alex Osborn popularized structured brainstorming in his 1953 book *Applied Imagination*.
6. **[Q6: Option B]** "Yes, and..." encourages additive, constructive idea building.
7. **[Q7: Option B]** Brainwriting allows silent, parallel idea generation without social intimidation.
8. **[Q8: Option B]** In SCAMPER, S stands for Substitute.
9. **[Q9: Option B]** Replacing one medium (paper) with another (digital QR code) is Substitution.
10. **[Q10: Option A]** Merging two distinct tools into a single entity is Combination.
11. **[Q11: Option C]** In SCAMPER, E stands for Eliminate.
12. **[Q12: Option B]** Inverting who performs the scanning (customer instead of cashier) is Reverse/Rearrange.
13. **[Q13: Option B]** Mind maps radiate organically from a central node through branching concepts.
14. **[Q14: Option B]** Reverse thinking explores how to cause a failure and inverts the findings for solutions.
15. **[Q15: Option A]** Affinity diagramming synthesizes messy data into thematic clusters.
16. **[Q16: Option B]** Concept bundling groups mutually reinforcing features into a single solution package.

17. [Q17: Option B] A coarse filter weeds out non-starters, followed by detailed multi-criteria matrix scoring.
18. [Q18: Option B] Calculation: $(0.6 \times 5) + (0.4 \times 2) = 3.0 + 0.8 = 3.8$.
19. [Q19: Option B] High Impact + Low Effort = Quick Wins (highest initial return for effort).
20. [Q20: Option C] High Effort + Low Impact ideas are "Money Pits" / "Thankless Tasks" and should be avoided.
21. [Q21: Option A] Dot voting provides a fast, democratic visual heat-map of team preferences.
22. [Q22: Option C] In the Double Diamond, Develop is the divergent phase for solution generation.
23. [Q23: Option B] First ideas are almost always superficial and derived from existing paradigms.
24. [Q24: Option B] The facilitator maintains process discipline, timekeeping, and psychological safety.
25. [Q25: Option B] Removing an existing component or feature is an application of "Eliminate".
26. [Q26: Option B] Originality reflects how novel, unique, and non-standard an idea is.
27. [Q27: Option B] Validating user need (desirability) ensures technical effort is not wasted.
28. [Q28: Option B] An Idea Backlog/Parking Lot preserves unselected ideas for future releases.
29. [Q29: Option B] Production blocking occurs when people must wait their turn to speak and lose ideas.
30. [Q30: Option B] Brainwriting allows everyone to write simultaneously in silence, eliminating blocking.

CHAPTER 4

Unit 4: Product Design and Prototyping

Practice Question Set (30 MCQs)

Q.1 What is the core definition of a prototype in Design Thinking?

- (A) A finished product ready for mass commercial distribution
- (B) A testable representation of an idea designed to answer specific questions and test assumptions
- (C) A legal patent document
- (D) A financial balance sheet

Q.2 What does the prototyping philosophy "Fail early, fail cheaply" mean?

- (A) Intentionally destroying products to claim insurance
- (B) Uncovering critical design flaws in low-cost prototypes before expensive production begins
- (C) Giving up on projects at the first sign of trouble
- (D) Lowering product prices below production costs

Q.3 Which of the following is a characteristic of a LOW-FIDELITY prototype?

- (A) Real-time database connectivity
- (B) Hand-drawn paper sketches, cardboard models, and rough mockups
- (C) High-resolution animations and pixel-perfect branding
- (D) Production-grade materials

Q.4 When should a design team primarily choose a Low-Fidelity prototype?

- (A) When performing final regulatory compliance testing
- (B) Early in the design process to explore concepts, layout, and user flows rapidly
- (C) When measuring microsecond software server response latency
- (D) For investor demonstrations of fully functional algorithms

Q.5 What is a key advantage of High-Fidelity clickable prototypes?

- (A) They can be created in 30 seconds with a pen
- (B) They allow realistic usability testing of detailed interactions and visual hierarchy
- (C) They require zero software tools
- (D) They prevent users from giving any negative feedback

Q.6 In a paper prototyping testing session, what role does the "Computer" play?

- (A) Records video of the test on a laptop
- (B) A facilitator who silently swaps paper screens and pop-ups in response to user taps
- (C) Analyzes statistical metrics automatically
- (D) Explains the backend database code to the participant

Q.7 What is a "Storyboard" in user experience design?

- (A) A screenplay written for a movie studio
- (B) A comic-strip style sequence of visual frames showing a user interacting with a product in real-world context
- (C) A task board showing software development sprint tickets
- (D) A list of customer testimonials

Q.8 What is the primary purpose of a Wireframe?

- (A) Demonstrating color palettes and typography branding
- (B) Providing a grayscale skeletal layout that establishes content hierarchy and interface structure
- (C) Showing 3D electrical wiring diagrams of a building
- (D) Writing automated database queries

Q.9 Why are wireframes traditionally created in grayscale (black, white, gray)?

- (A) Because color printing is too expensive
- (B) To keep focus strictly on layout, information hierarchy, and functionality without color distraction
- (C) Because computers cannot display color during early design
- (D) Because users dislike colorful interfaces

Q.10 In UI/UX wireframing conventions, what does a rectangle with an 'X' drawn across it represent?

- (A) A delete button
- (B) An image or media placeholder
- (C) A broken database link
- (D) A closed window

Q.11 When should Radio Buttons be used in an interface wireframe?

- (A) When a user can select multiple options simultaneously
- (B) When a user must choose exactly ONE option from a mutually exclusive list
- (C) For scrolling through long pages
- (D) For entering password text

Q.12 When should Checkboxes be used in a form?

- (A) When a user may select zero, one, or MULTIPLE independent options
- (B) When only one binary choice is allowed
- (C) To trigger an immediate payment transfer
- (D) To submit a form to the server

Q.13 In a user flow diagram, what does an Oval (Terminator) shape represent?

- (A) A decision point
- (B) The Start or End of a user workflow
- (C) A database storage unit
- (D) A clickable button

Q.14 In a user flow diagram, what does a Diamond shape represent?

- (A) A user screen
- (B) A conditional decision point with branching paths (e.g., Yes / No)
- (C) The start of the process
- (D) A payment gateway

Q.15 What is the "Happy Path" (Primary Path) in a user flow?

- (A) A colorful design theme
- (B) The default, error-free path where a user accomplishes their goal in the shortest direct sequence
- (C) A path that only children use
- (D) A marketing advertisement flow

Q.16 What is a Minimum Viable Product (MVP)?

- (A) A stripped-down, broken version of a product that barely functions
- (B) The smallest version of a product that delivers real value to early adopters and enables maximum validated learning
- (C) A product with every single conceivable feature included
- (D) A product built with zero budget

Q.17 Which of the following is a classic analogy for an MVP?

- (A) Delivering a single wheel, then an axle, then a car chassis
- (B) Delivering a skateboard first, then a bicycle, then a motorcycle, then a car (each stage is functional transport)
- (C) Delivering a car without brakes
- (D) Delivering an engine sitting on the floor

Q.18 What is a "Landing Page MVP"?

- (A) An airport runway management software
- (B) A single web page describing a value proposition with a "Sign Up" button to gauge consumer demand before building
- (C) A fully coded e-commerce website with 50,000 products
- (D) A printed brochure handed out on the street

Q.19 What is a "Concierge MVP"?

- (A) A software product for luxury hotel receptionists
- (B) Delivering the product service manually and personally behind the scenes to learn user needs before automating
- (C) An automated AI chatbot
- (D) A physical prototype made of titanium

Q.20 What is a "Wizard of Oz" prototype?

- (A) A fantasy video game prototype
- (B) A prototype that looks completely automated to the front-end user but has humans manually operating backend tasks
- (C) A prototype that changes color dynamically
- (D) A prototype built using voice-over actors

Q.21 Why is testing with representative content (realistic text/prices) superior to "Lorem Ipsum" in wireframes?

- (A) Lorem Ipsum is illegal in software design
- (B) Realistic content exposes actual spacing constraints, text overflow, and cognitive comprehension issues
- (C) Realistic text makes the wireframe look like a finished website
- (D) It prevents copyright violations

Q.22 Which prototype fidelity is best suited for testing complex micro-interactions, like swipe gestures and animated transitions?

- (A) Paper sketch
- (B) Cardboard cutout
- (C) High-Fidelity interactive digital prototype
- (D) Static text document

Q.23 In digital product design, what does an "Empty State" design represent?

- (A) A screen when the user's internet is disconnected

- (B) What a user sees when there is no data to display yet (e.g., "No orders placed yet"), guiding them to take action
- (C) A completely blank white screen due to a server crash
- (D) A logout confirmation screen

Q.24 What is the riskiest assumption for a startup building an on-demand laundry app?

- (A) What shade of blue to use for the logo
- (B) Whether customers are willing to trust strangers to wash and return their clothes reliably
- (C) Whether to use React or Angular for the web frontend
- (D) The font size of the copyright notice

Q.25 How many user testing participants are typically recommended by usability expert Jakob Nielsen for discovering 85% of usability errors in a prototype?

- (A) 100 users
- (B) Exactly 5 users per target user group
- (C) 1,000 users
- (D) Only 1 user

Q.26 What does "Traceability" mean in product design?

- (A) Tracing drawings on transparent paper
- (B) The ability to track every wireframe component and feature back to a validated user need and requirement
- (C) Recording user GPS locations
- (D) Following competitors' design styles

Q.27 In wireframing, what is the role of an "Annotation"?

- (A) Artistic decoration in the margins
- (B) Numbered descriptive notes explaining dynamic behaviors, validation rules, and error states
- (C) User comments on social media
- (D) Software pricing details

Q.28 Which of the following is an example of an Alternate Path in a checkout user flow?

- (A) User enters address → enters card → payment succeeds
- (B) User's credit card fails → system prompts for another payment method → user retries
- (C) User searches product → views product
- (D) User opens app → logs in

Q.29 What is the main danger of building a High-Fidelity prototype too early in a project?

- (A) It uses too much electricity
- (B) Teams become emotionally attached to the visual polish and resist making fundamental structural changes
- (C) Users refuse to test beautiful prototypes
- (D) High-fidelity prototypes cannot be shared online

Q.30 What constitutes a complete, testable MVP hypothesis?

- (A) "We believe [target user] has [problem], and will achieve [outcome] using [feature/solution], measured by [metric]."
- (B) "We believe our app will make millions of dollars by next month."
- (C) "We believe everyone in the world needs our new software."
- (D) "We believe technology is the future of business."

Answer Key & Step-by-Step Explanations

Unit 4: Quick Answer Key

Q1 B	Q2 B	Q3 B	Q4 B	Q5 B	Q6 B	Q7 B	Q8 B	Q9 B	Q10 B
Q11 B	Q12 A	Q13 B	Q14 B	Q15 B	Q16 B	Q17 B	Q18 B	Q19 B	Q20 B
Q21 B	Q22 C	Q23 B	Q24 B	Q25 B	Q26 B	Q27 B	Q28 B	Q29 B	Q30 A

Detailed Pedagogical Explanations

1. **[Q1: Option B]** A prototype is an exploratory learning tool to test specific assumptions.
2. **[Q2: Option B]** Early cheap failures reveal wrong assumptions before major capital is committed.
3. **[Q3: Option B]** Lo-Fi prototypes are quick, rough representations like paper sketches and cardboard.
4. **[Q4: Option B]** Lo-Fi is ideal early on for exploring structures, workflows, and core concepts.
5. **[Q5: Option B]** Hi-Fi prototypes provide realistic interaction for testing fine UI and usability details.
6. **[Q6: Option B]** In paper prototyping, the "Computer" physically manipulates paper states based on user actions.
7. **[Q7: Option B]** A storyboard visually illustrates the user journey and context across comic-like panels.
8. **[Q8: Option B]** Wireframes establish structural layout, visual hierarchy, and interface functional zones.
9. **[Q9: Option B]** Grayscale keeps feedback focused on structure and function rather than subjective color debates.
10. **[Q10: Option B]** A crossed rectangle is the universal design blueprint placeholder for images/video.
11. **[Q11: Option B]** Radio buttons represent single-choice mutually exclusive selections.
12. **[Q12: Option A]** Checkboxes allow zero, one, or multiple independent selections.
13. **[Q13: Option B]** Ovals (terminators) denote the starting and ending points of a flowchart.
14. **[Q14: Option B]** Diamond shapes represent conditional decision logic (e.g., Is user logged in?).
15. **[Q15: Option B]** The happy path is the primary frictionless route to task completion.

16. **[Q16: Option B]** An MVP is the leanest functional product that delivers core value and generates learning.
17. **[Q17: Option B]** An MVP must provide functional value at every stage (skateboard → car).
18. **[Q18: Option B]** Landing page MVPs test market demand and click interest before writing code.
19. **[Q19: Option B]** Concierge MVPs deliver services manually to learn direct customer needs.
20. **[Q20: Option B]** Wizard of Oz prototypes simulate automated backends with hidden manual human labor.
21. **[Q21: Option B]** Real content exposes actual text wrapping, button sizing, and cognitive clarity.
22. **[Q22: Option C]** Hi-Fi digital prototypes are required to test animated transitions and responsive gestures.
23. **[Q23: Option B]** Empty states guide first-time users when no data exists yet.
24. **[Q24: Option B]** Customer trust in handing over laundry is the fundamental value assumption.
25. **[Q25: Option B]** Jakob Nielsen's research showed 5 users uncover 85% of usability issues.
26. **[Q26: Option B]** Traceability connects UI components directly back to user requirements.
27. **[Q27: Option B]** Annotations document dynamic behavior, data validation, and logic for developers.
28. **[Q28: Option B]** Payment retry/recovery represents an alternate/exception flow path.
29. **[Q29: Option B]** High early polish causes sunk cost fallacy and attachment to flawed structures.
30. **[Q30: Option A]** A structured hypothesis links user, problem, proposed solution, and measurable metric.

CHAPTER 5

Unit 5: Testing, Validation and Customer Experience

Practice Question Set (30 MCQs)

Q.1 What is the primary purpose of user testing in Design Thinking?

- (A) To prove to management that the design is flawless
- (B) To observe real user behavior and identify usability defects, confusion, and invalid assumptions
- (C) To grade the intelligence of the user
- (D) To sell subscriptions to test participants

Q.2 What is the fundamental difference between Usability Testing and Design Validation?

- (A) Usability testing checks if the interface works efficiently; validation checks if it solves the right user problem
- (B) Usability testing is done by developers; validation is done by accountants
- (C) Usability testing requires paper; validation requires code
- (D) There is no difference; they are exact synonyms

Q.3 Which of the following best exemplifies the "Show, Don't Tell" testing approach?

- (A) Presenting a 30-minute PowerPoint pitch explaining why the app is easy to use
- (B) Handing the user a prototype with a task scenario and observing their actions silently
- (C) Writing an instruction manual and reading it aloud to the user
- (D) Showing a promotional video of the product

Q.4 During a neutral user testing session, what should the facilitator say when a user asks, "Should I click this button?"

- (A) "Yes, that is the correct button!"
- (B) "What do you think will happen if you click it?"
- (C) "No, don't click that, it's wrong."
- (D) "Let me click it for you."

Q.5 Why is it important to tell the participant: "We are testing the product, not you"?

- (A) To protect the company from legal liability
- (B) To reduce participant anxiety and ensure they do not feel at fault when design errors occur
- (C) To prevent the participant from asking for money
- (D) Because participants are not allowed to make mistakes

Q.6 What makes a user testing Task Scenario effective?

- (A) It specifies every button and menu name in exact order
- (B) It provides a realistic goal and context without dictating specific interface steps
- (C) It has no defined end goal
- (D) It can be completed in less than one second

Q.7 Which of the following is a POORLY constructed task scenario?

- (A) "Find a pair of running shoes under Rs. 2,000 and proceed to the shipping screen."
- (B) "Click the search bar, type 'shoes', click filter, select 'under 2000', and tap 'Add to Cart'."
- (C) "You need to book a flight to Mumbai for next Monday morning. Show how you would do that."
- (D) "Check your account balance and download last month's statement."

Q.8 What is the "Think-Aloud Protocol" in usability testing?

- (A) The facilitator constantly explains what the software is doing
- (B) The participant verbalizes their thoughts, expectations, and hesitations aloud while performing the task
- (C) All observers in the room shout feedback at the participant
- (D) The participant recites the user manual from memory

Q.9 Which quadrant of the Feedback Capture Matrix records constructive criticisms and unmet user wishes?

- (A) Likes (+)
- (B) Wishes (Δ)
- (C) Questions (?)
- (D) Ideas (!)

Q.10 In a Feedback Capture Matrix, where does a user's statement "Is international shipping included in this price?" belong?

- (A) Likes (+)
- (B) Wishes (Δ)
- (C) Questions (?)
- (D) Ideas (!)

Q.11 How does Customer Experience (CX) differ from Usability (UI/UX)?

- (A) Usability is broader than CX
- (B) Usability evaluates specific interface interactions; CX encompasses the holistic end-to-end relationship across all touchpoints
- (C) Usability only applies to websites; CX only applies to physical retail stores

(D) Usability is measured in revenue; CX is measured in clicks

Q.12 Which of the following is an example of a HUMAN touchpoint in a service journey?

- (A) An automated billing email
- (B) A delivery courier handing a package to the customer
- (C) The website homepage
- (D) The cardboard product packaging

Q.13 In Customer Journey Mapping, what are the primary layers mapped across stages?

- (A) User Actions, Thoughts/Emotions, Touchpoints, Pain Points, and Opportunities
- (B) HTML code, CSS stylesheets, JavaScript files, and Server logs
- (C) Profit margins, Depreciation rates, and Tax brackets
- (D) Employee salaries and Office desk locations

Q.14 What does the "P" stand for in the WCAG accessibility POUR principles?

- (A) Predictable
- (B) Perceivable
- (C) Portable
- (D) Practical

Q.15 According to WCAG 2.1 guidelines, what is the minimum color contrast ratio required for standard body text (Level AA)?

- (A) 1.5:1
- (B) 3.0:1
- (C) 4.5:1
- (D) 10:1

Q.16 What accessibility feature ensures that visually impaired users using screen readers can understand images?

- (A) High-resolution 4K images
- (B) Descriptive Alternative Text ('alt' attribute)
- (C) Flashing neon borders
- (D) Background audio music

Q.17 Which of the following violates the "Operable" principle of web accessibility?

- (A) An interface that can be fully navigated using only a keyboard
- (B) A modal dialog that creates a "Keyboard Trap" from which keyboard-only users cannot escape
- (C) Providing visible focus outlines on active input fields

(D) Allowing users to pause auto-rotating carousels

Q.18 What is a "Breadcrumb" in digital software navigation?

(A) A cookie saved on a hard drive

(B) A secondary navigation trail showing the user's current location within a website hierarchy (e.g., Home > Electronics > Audio)

(C) A temporary error message

(D) An advertisement popup

Q.19 What does the Usability heuristic "Visibility of System Status" mean?

(A) The software must display its server IP address

(B) The system should always keep users informed about what is going on through appropriate, timely feedback (e.g., loading bars)

(C) Making the entire source code open source

(D) Showing employee names on the homepage

Q.20 When an error occurs on a form, what is the best UX practice for displaying error messages?

(A) Show a cryptic error code like "Error 0x800F" at the top of the page

(B) Display a clear, friendly text message placed directly adjacent to the offending input field explaining how to fix it

(C) Clear the entire form and force the user to re-enter all fields

(D) Hide the submit button permanently

Q.21 Why should destructive actions (like "Delete Account" or "Format Disk") always require explicit confirmation?

(A) To slow down the user intentionally

(B) To prevent catastrophic irreversible errors caused by accidental clicks

(C) To increase database server load

(D) Because computer code cannot delete data directly

Q.22 What metric measures the percentage of users who successfully finish a designated test task without moderator help?

(A) Task Completion Rate (Success Rate)

(B) Net Promoter Score

(C) Bounce Rate

(D) Return on Investment

Q.23 Which metric evaluates customer loyalty by asking "How likely are you to recommend this product to a friend or colleague?" on a 0–10 scale?

- (A) Customer Effort Score (CES)
- (B) Net Promoter Score (NPS)
- (C) System Usability Scale (SUS)
- (D) Task Completion Time

Q.24 What is a "Single Ease Question" (SEQ) in usability testing?

- (A) A 50-question survey given before the test
- (B) A single 7-point rating scale question asked immediately after a task: *"Overall, how easy or difficult was this task?"*
- (C) A riddle asked by the interviewer
- (D) A multiple-choice exam question

Q.25 What is the main limitation of automated accessibility testing tools (like Lighthouse or Axe)?

- (A) They are too expensive to run
- (B) They can only detect 30–40% of accessibility issues automatically (cannot judge text clarity, alt-text quality, or logical flow)
- (C) They destroy web page code
- (D) They only work on mobile phones

Q.26 If a customer loves an airline's mobile booking app but has their luggage lost by baggage handlers, how is the overall Customer Experience (CX) affected?

- (A) CX remains 100% positive because the app had good usability
- (B) CX is severely degraded because CX reflects the holistic end-to-end journey across digital and physical touchpoints
- (C) CX only measures mobile software interactions
- (D) The airline's Net Promoter Score will automatically rise

Q.27 In digital navigation, what does "Recognition rather than Recall" mean?

- (A) Forcing users to memorize menu paths from previous pages
- (B) Making options, actions, and objects visible so users do not have to remember information across screens
- (C) Using facial recognition to log in
- (D) Asking users to type exact command-line codes

Q.28 What type of touchpoint is a monthly paper electricity bill delivered to a customer's mailbox?

- (A) Digital touchpoint
- (B) Physical touchpoint
- (C) Human touchpoint
- (D) Social touchpoint

Q.29 Why should user testing include participants with diverse abilities and accessibility requirements?

- (A) It is only required if the government threatens legal action
- (B) Inclusive design uncovers barriers early and creates products that are easier and better for everyone
- (C) Accessible testing reduces the need for software code
- (D) Disabled users only test hardware devices

Q.30 What should a design team do immediately after completing a round of user tests?

- (A) Celebrate and launch the product to market without changes
- (B) Synthesize observations, populate a Feedback Capture Matrix, identify critical themes, and iterate the prototype
- (C) Delete the user recordings
- (D) Fire the developers whose features failed

Answer Key & Step-by-Step Explanations

Unit 5: Quick Answer Key

Q1 B	Q2 A	Q3 B	Q4 B	Q5 B	Q6 B	Q7 B	Q8 B	Q9 B	Q10 C
Q11 B	Q12 B	Q13 A	Q14 B	Q15 C	Q16 B	Q17 B	Q18 B	Q19 B	Q20 B
Q21 B	Q22 A	Q23 B	Q24 B	Q25 B	Q26 B	Q27 B	Q28 B	Q29 B	Q30 B

Detailed Pedagogical Explanations

1. **[Q1: Option B]** Testing is an empirical learning tool to discover flaws and validate assumptions with real users.
2. **[Q2: Option A]** Usability checks functional interaction quality; validation checks problem-solution alignment.
3. **[Q3: Option B]** "Show, Don't Tell" relies on direct user interaction and neutral observation rather than sales pitches.
4. **[Q4: Option B]** Neutral probes encourage the user to verbalize their mental model without coaching.
5. **[Q5: Option B]** Reassuring participants ensures they don't feel self-conscious or blame themselves for UI failures.
6. **[Q6: Option B]** Effective scenarios give realistic goals and context without prescribing button clicks.
7. **[Q7: Option B]** Specifying every exact UI step tests obedience, not discoverability or usability.
8. **[Q8: Option B]** The think-aloud protocol captures the participant's internal thoughts and expectations live.
9. **[Q9: Option B]** The Wishes (Δ) quadrant captures constructive criticisms and unmet needs.
10. **[Q10: Option C]** Unanswered questions and user confusion belong in the Questions (?) quadrant.
11. **[Q11: Option B]** CX is the overarching journey across all touchpoints; usability is product-specific interaction.
12. **[Q12: Option B]** Delivery couriers, support reps, and retail staff are human service touchpoints.
13. **[Q13: Option A]** Journey maps systematically track Actions, Thoughts/Emotions, Touchpoints, Pain Points, and Opportunities.
14. **[Q14: Option B]** POUR stands for Perceivable, Operable, Understandable, Robust.

15. **[Q15: Option C]** WCAG AA mandates at least a 4.5:1 contrast ratio for standard text.
16. **[Q16: Option B]** Screen readers read descriptive alternative text ('alt' attribute) aloud for non-visual users.
17. **[Q17: Option B]** Keyboard traps trap focus, preventing keyboard-only users from navigating the site.
18. **[Q18: Option B]** Breadcrumbs provide spatial orientation and easy navigation back up the hierarchy.
19. **[Q19: Option B]** Systems must provide clear and prompt feedback regarding ongoing operations.
20. **[Q20: Option B]** Inline error messages adjacent to fields explain the exact issue and recovery step.
21. **[Q21: Option B]** Confirmations prevent destructive, irreversible accidents.
22. **[Q22: Option A]** Task Completion Rate measures the proportion of successfully achieved goals.
23. **[Q23: Option B]** Net Promoter Score measures advocacy and loyalty on an 11-point scale (0–10).
24. **[Q24: Option B]** Single Ease Question is a standardized post-task ease rating.
25. **[Q25: Option B]** Automated tools catch code syntax but cannot judge human cognitive clarity or task flow.
26. **[Q26: Option B]** CX is holistic; a breakdown in physical service destroys overall customer trust.
27. **[Q27: Option B]** Minimizing user memory load by making elements visible promotes usability.
28. **[Q28: Option B]** Printed bills, packaging, and hardware devices are physical touchpoints.
29. **[Q29: Option B]** Inclusive design solves for one and extends to many, creating universal ease of use.
30. **[Q30: Option B]** Immediate structured synthesis and iteration turn test observations into product improvements.

CHAPTER 6

Unit 6: Innovation Project, Re-Design and Product Presentation

Practice Question Set (30 MCQs)

Q.1 What is "Design Iteration"?

- (A) Copying an existing design without modification
- (B) The cyclical process of prototyping, testing, analyzing feedback, and refining a solution
- (C) Writing code once and never modifying it
- (D) Designing only on alternate days of the week

Q.2 What is the fundamental difference between Design Refactoring and Product Re-Design?

- (A) Refactoring changes the logo color; Re-design deletes the project
- (B) Refactoring improves internal structure and consistency without changing core function; Re-design fundamentally alters structure, form, or workflow
- (C) Refactoring is done by designers; Re-design is done by lawyers
- (D) There is no difference between refactoring and re-design

Q.3 When should a team undertake a fundamental Product Re-Design rather than minor iteration?

- (A) When a single user dislikes a font
- (B) When testing reveals that core problem assumptions, basic architecture, or safety mechanisms have failed
- (C) When the sprint deadline is 2 hours away
- (D) When the marketing team wants a new slogan

Q.4 In Lean Startup methodology, what is a "Strategic Pivot"?

- (A) Rotating a 3D model in CAD software
- (B) A structured course correction designed to test a new fundamental hypothesis about the product, customer segment, or business model while keeping validated learnings
- (C) Firing the entire executive board
- (D) Shutting down the company permanently

Q.5 Which of the following is a classic example of a "Customer Segment Pivot"?

- (A) Rewriting a Python backend in Rust

- (B) Shifting a product originally built for individual college students to enterprise corporate HR departments
- (C) Changing the company office address
- (D) Adding dark mode to a mobile app

Q.6 What does the formula $P = F \times S \times C$ calculate in user feedback prioritization?

- (A) Product sales profit
- (B) Priority score based on Frequency, Severity, and Confidence in evidence
- (C) Software processing speed
- (D) Project manager salary

Q.7 In user feedback prioritization, why does a rare safety failure (High Severity, Low Frequency) often outrank a common cosmetic glitch (Low Severity, High Frequency)?

- (A) Cosmetic issues cannot be fixed
- (B) Severe safety or task-blocking failures carry catastrophic risk and must be remediated immediately
- (C) Frequency is the only metric that matters in engineering
- (D) Severity is ignored in modern product design

Q.8 What is "Regression Testing" during design refactoring and product iteration?

- (A) Testing whether older users can operate the app
- (B) Verifying that recent improvements or changes have not unintentionally broken previously working functionality
- (C) Testing software on obsolete computers
- (D) Reverting to the very first sketch

Q.9 What are the four stages of the continuous improvement PDCA cycle?

- (A) Produce, Deliver, Calculate, Archive
- (B) Plan, Do, Check, Act
- (C) Pitch, Demonstrate, Close, Acquire
- (D) Propose, Design, Construct, Authorize

Q.10 What is the primary rule of Audience-Centred Product Presentation?

- (A) Use the exact same technical jargon regardless of who is listening
- (B) Tailor the language, narrative focus, and evidence to the specific priorities of the audience (investors, users, or engineers)
- (C) Speak as fast as possible to fit all information into the slides
- (D) Avoid showing the prototype to the audience

Q.11 When presenting a product to executive business investors, the presentation should focus most on:

- (A) Transistor gate schematics and code line counts
- (B) Market viability, unit economics, customer acquisition, scalability, and ROI
- (C) CSS stylesheet class names
- (D) The personal biographies of all junior designers

Q.12 What does the acronym NABC stand for in innovation pitching?

- (A) Network, Action, Budget, Capital
- (B) Need, Approach, Benefits per costs, Competition / alternatives
- (C) Novelty, Architecture, Branding, Customer
- (D) Narrative, Audience, Benchmark, Conclusion

Q.13 In the NABC framework, what does the "Need" section address?

- (A) The team's need for funding
- (B) A specific, validated user pain point backed by real-world evidence
- (C) The server hardware requirements
- (D) The company's legal registration

Q.14 In the NABC framework, why is it critical to explicitly analyze "Competition and Alternatives"?

- (A) To mock competitors publicly
- (B) To demonstrate how your solution provides superior relative advantage compared to existing workarounds, competitors, and non-consumption
- (C) Because investors require buying competitor shares
- (D) To copy competitors' marketing slogans

Q.15 What is an "Elevator Pitch"?

- (A) A presentation delivered inside a moving elevator
- (B) A concise 30–60 second verbal summary that clearly explains the problem, solution, unique value, and a requested call-to-action
- (C) A 2-hour keynote address
- (D) A written technical dissertation

Q.16 What is the ideal duration of a standard elevator pitch?

- (A) 10 to 15 seconds
- (B) 30 to 60 seconds
- (C) 15 to 20 minutes

(D) Exactly 1 hour

Q.17 Which of the following represents the most compelling opening "Hook" for a product presentation?

(A) "Hello, our names are on slide 1, and today we will talk about our project."

(B) "Every year, 40% of critical hospital medicine spoils due to undetected temperature spikes in transit. Here is how we solve that."

(C) "We spent 100 hours writing 5,000 lines of Python code."

(D) "Our company was founded in 2026."

Q.18 What is the "Rule of One" in high-impact presentation slide design?

(A) Every presentation must only have one single slide

(B) Each slide should focus on conveying ONE core idea with clean visual support

(C) Only one person in the audience is allowed to ask questions

(D) Use only one color across all slides

Q.19 Why should product demonstrations focus on a "Representative User Task" rather than showing every single setting?

(A) Setting menus are confidential

(B) Demonstrating an authentic end-to-end task proves how the product delivers core value in real context

(C) It prevents the computer from crashing

(D) Users never use settings menus

Q.20 What is a "Design Rationale"?

(A) An excuse made when a project fails

(B) The explicit justification and evidence-based reasoning behind why a specific design choice was made

(C) The total monetary cost of the design

(D) The list of software libraries used

Q.21 Why is stating known limitations and failure conditions important during a final technical case study presentation?

(A) It makes the team look incompetent

(B) It demonstrates technical integrity, thorough testing, and builds strong credibility with evaluators

(C) Limitations are required by criminal law

(D) It guarantees immediate venture capital funding

Q.22 In a design case study, what does "Requirement Traceability" demonstrate?

- (A) That code was written by tracing paper blueprints
- (B) A clear, unbroken chain connecting user needs → engineering specifications → prototype features → verification test results
- (C) Tracking the GPS coordinates of team members
- (D) Calculating software download speeds

Q.23 When giving a live prototype demonstration, why is having a pre-recorded backup video considered essential?

- (A) Because live prototypes are illegal in formal presentations
- (B) To safeguard against unexpected live network, hardware, or environmental glitches during the pitch
- (C) So the presenter can leave the room early
- (D) Because video editing is mandatory for grading

Q.24 Which elevator pitch template is widely considered standard in product management?

- (A) "For [target user] who [problem], our [product] is a [category] that [benefit]. Unlike [competition], we [differentiator]."
- (B) "Buy our product because it is the cheapest and best in the world."
- (C) "We have built a mobile app that does everything for everyone."
- (D) "Our product has AI and blockchain built inside."

Q.25 What is the danger of relying on "Feature Bloat" (adding endless features) to save a struggling product?

- (A) It increases software file size only
- (B) It adds cognitive clutter, increases maintenance costs, and fails to fix an unvalidated core value proposition
- (C) It makes the app too easy to use
- (D) Feature bloat is always recommended by designers

Q.26 What does the "Benefits per Costs" section in the NABC model evaluate?

- (A) The retail price of the prototype box
- (B) The quantitative and qualitative value delivered compared to the financial, time, and adoption effort required by the user
- (C) The monthly salary of the engineers
- (D) The cost of office furniture

Q.27 During a final team product presentation, what is the best practice for team delivery?

- (A) Have one person talk for 100% of the time while others look at their phones
- (B) Coordinate defined speaker transitions, maintain consistent terminology, and show shared ownership of the project

- (C) Argue with teammates on stage about technical details
- (D) Read directly from full paragraphs of text on the slides

Q.28 What should immediately follow the conclusion of an effective product pitch?

- (A) A clear, specific "Ask" or Call-to-Action (e.g., requesting pilot testing access, partnership, or seed funding)
- (B) Leaving the stage in total silence
- (C) Apologizing for any mistakes made during the presentation
- (D) Handing out paper business cards without speaking

Q.29 What type of strategic pivot occurred when YouTube pivoted from a video dating platform to a universal video hosting platform?

- (A) Technology architecture pivot
- (B) Problem / Value proposition & customer segment pivot
- (C) Zoom-in pivot
- (D) Channel pivot

Q.30 What constitutes the final comprehensive deliverable of an INT335 Design Thinking innovation project?

- (A) A single PowerPoint slide with a logo
- (B) An evidence-backed portfolio including research synthesis, iterative prototypes, test logs, technical specs, and a persuasive demonstration
- (C) A 100-page theoretical essay with no prototype
- (D) A list of marketing slogans

Answer Key & Step-by-Step Explanations

Unit 6: Quick Answer Key

Q1 B	Q2 B	Q3 B	Q4 B	Q5 B	Q6 B	Q7 B	Q8 B	Q9 B	Q10 B
Q11 B	Q12 B	Q13 B	Q14 B	Q15 B	Q16 B	Q17 B	Q18 B	Q19 B	Q20 B
Q21 B	Q22 B	Q23 B	Q24 A	Q25 B	Q26 B	Q27 B	Q28 A	Q29 B	Q30 B

Detailed Pedagogical Explanations

1. **[Q1: Option B]** Iteration is the continuous learning loop of building, testing, learning, and refining.
2. **[Q2: Option B]** Refactoring improves internal quality/structure; re-design overhauls the fundamental solution.
3. **[Q3: Option B]** Invalidation of foundational assumptions or critical safety/architectural flaws mandates re-design.
4. **[Q4: Option B]** A pivot is a structured hypothesis pivot that tests a new strategy while preserving past learning.
5. **[Q5: Option B]** Moving from B2C students to B2B enterprise clients is a customer segment pivot.
6. **[Q6: Option B]** $P = F \times S \times C$ ranks feedback by Frequency, Severity, and Confidence.
7. **[Q7: Option B]** High-severity failures can cause total task abandonment or safety disasters.
8. **[Q8: Option B]** Regression testing ensures new code or design changes do not break existing working flows.
9. **[Q9: Option B]** The continuous improvement cycle is Plan → Do → Check → Act.
10. **[Q10: Option B]** Effective communication aligns message and evidence directly with audience interests.
11. **[Q11: Option B]** Investors focus on market viability, business sustainability, unit economics, and return.
12. **[Q12: Option B]** NABC stands for Need, Approach, Benefits per costs, and Competition/alternatives.
13. **[Q13: Option B]** "Need" defines the genuine, validated user pain point backed by research evidence.
14. **[Q14: Option B]** Comparing with alternatives highlights the unique relative advantage of your innovation.

15. **[Q15: Option B]** An elevator pitch delivers a high-impact product summary within 30–60 seconds.
16. **[Q16: Option B]** Standard elevator pitches are strictly timeboxed between 30 and 60 seconds.
17. **[Q17: Option B]** A compelling hook uses a vivid, quantified problem statement to command immediate attention.
18. **[Q18: Option B]** The "Rule of One" limits each slide to one dominant idea to maximize audience retention.
19. **[Q19: Option B]** Demonstrating a realistic user task proves the solution's real-world value proposition.
20. **[Q20: Option B]** Design rationale provides the empirical evidence justifying why specific choices were made.
21. **[Q21: Option B]** Acknowledging limitations demonstrates rigor, maturity, and builds evaluator trust.
22. **[Q22: Option B]** Traceability connects user needs through engineering specs to test verification results.
23. **[Q23: Option B]** A backup demo video protects against live hardware, venue Wi-Fi, or software crashes.
24. **[Q24: Option A]** The standard positioning template clearly specifies user, problem, category, benefit, and differentiator.
25. **[Q25: Option B]** Feature bloat increases interface complexity without addressing core usability or value problems.
26. **[Q26: Option B]** Benefits per costs assesses total net value relative to the customer's effort, price, and friction.
27. **[Q27: Option B]** Coordinated transitions and shared delivery reflect strong professional team cohesion.
28. **[Q28: Option A]** Every presentation must culminate in a concrete, actionable call-to-action.
29. **[Q29: Option B]** YouTube pivoted its core value proposition and audience from video dating to universal video sharing.
30. **[Q30: Option B]** A complete design project integrates research, prototypes, test logs, technical specs, and pitch.